

decision-making, making it a key focus in Neuro-Symbolic AI. Present research within Neuro-Symbolic AI does not yet effectively cover meta-cognition and neglecting Meta-Cognition in Neuro-Symbolic AI research limits system autonomy, adaptability, and reliability, hindering error correction and reducing trustworthiness in dynamic environments, making self-awareness and self-regulation essential for future advancements.

2.3. Literature Review Approach

We followed the PRISMA systematic review methodology to ensure a thorough and unbiased survey of the literature. Our search was conducted across five databases: [IEEE Explore](#), [Google Scholar](#), [arXiv](#), [ACM Digital Library](#), and [SpringerLink Library](#), focusing on publications from 2020 to 2024. The keywords included "Neuro-Symbolic" combined with terms related to the foundational research areas. Only peer-reviewed articles, conference papers, and books in English were considered. From an initial broad search, we refined our selection to 392 candidate papers, which were further screened for quality, relevance, and availability of a public codebase, resulting in 167 papers that were included in our review. This streamlined process allowed us to focus on the most relevant and high-quality research, identifying key findings and open problems in Neuro-Symbolic AI.

3. Results

Database	Knowledge	Learning & Inference	Explainability & Trustworthiness	Logic & Reasoning	Meta-Cognition
IEEE	73	97	15	67	33
Google Scholar	56	126	7	129	3
ArXiv	17	54	7	55	3
ACM	10	46	5	12	17
Springer	152	170	65	162	47
Total(after screening)	308	493	99	425	103

Table 1

The search terms "neurosymbolic" AND each of the terms required for the 5 foundational research areas within neurosymbolic AI were queried through the 5 databases. The number of pieces of literature returned from each query is shown in the table above. Note also that only publications from 2020-2024 were considered

From the initial Google Scholar scraping, there was a total of 957 publications listed on Google Scholar alone from 1970 til the present. [Figure 1](#) shows how research on Neuro-Symbolic AI is increasing exponentially starting in 2020, with notable increases in the years beginning from 2020 (53 publications), and peaking in 2023 (236 publications). Combining the Google Scholar literature from 2020 onwards with the literature from the four other databases queried for pieces of literature on Neuro-Symbolic AI from 2020 onwards, a total of 1,428 papers were extracted as illustrated in [table 1](#). An illustration of this sub-categorisation showing the overlap between 4 of the 5 main research focal areas is shown in [Figure 2](#). From the total literature extracted, 45% (n = 641) were removed as duplicate entries, and 28% (n = 395) were removed during title and

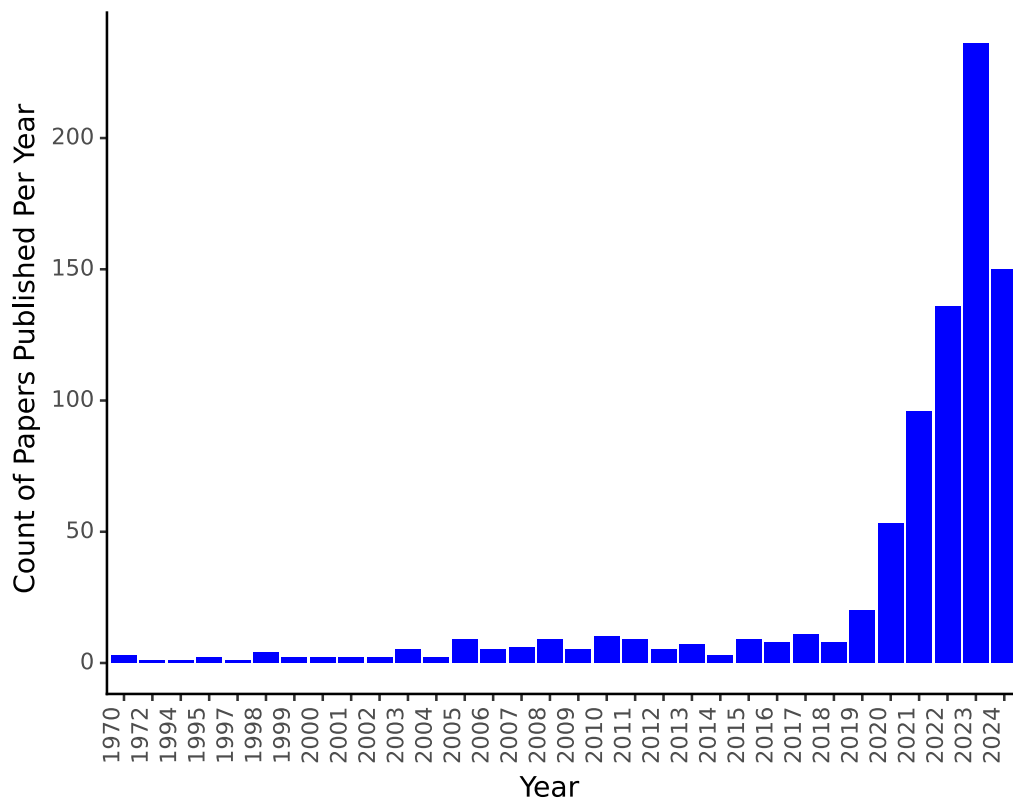


Figure 1: Histogram of publications per year for Neuro-Symbolic AI. The data was obtained through Google Scholar scraping, reflecting significant growth from 2020

abstract screening with 28% ($n = 392$) held for further analysis. From the remaining 392 papers, the literature was further split based on code/model availability. 42% of the papers ($n = 167$) had associated code-base repositories (e.g. GitHub, Huggingface etc.) and 58% ($n = 225$) were further excluded from this literature review as a public code-base could not be found for the associated piece of literature (except for entries on Meta-Cognition as no code-bases could be found for literature associated with this category). The remaining 167 papers gathered were then read in detail and a further 9 papers were removed for not meeting the inclusion criteria leaving 158 included papers and 234 excluded papers. These 158 papers were then sub-categorised under the five main focal research areas and the intersection found therein. There were 44% ($n=70$) entries in the Knowledge Representation category, 63% ($n=99$) entries in the Learning and Inference category, 28% ($n=44$) entries in the Explainability and Trustworthiness category, 35% ($n=55$) entries in the Logic and Reasoning category, 5% ($n=8$) entries in the Meta-Cognition category. The intersection of Knowledge Representation and Learning & Inference had 27% ($n=43$) entries, the intersection of Knowledge Representation and Explainability & Trustworthiness had 4% ($n=5$), the intersection of Learning & Inference and Logic & Reasoning had 11.48% ($n=31$), the

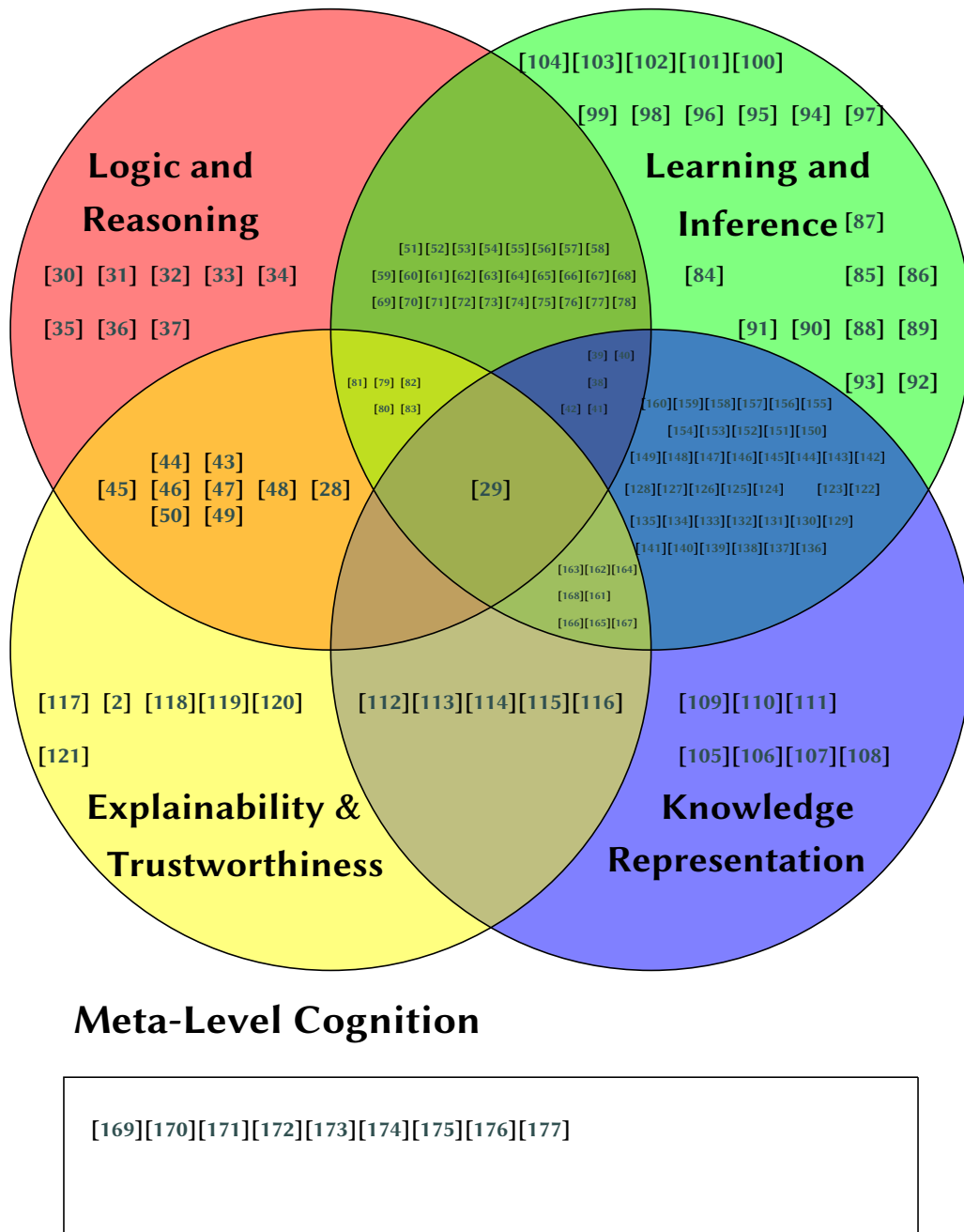


Figure 2: A literature review of existing of the major components of Symbolic AI was conducted. Note that papers from the Meta-Level Cognition were not required to have an associated public code-base/repository